

The Tensor-Power Scale in the Hypergraph Matrix Lemma Is Optimal

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Status: candidate full solution, likely valid, to the first question in Remark 2.3 of arXiv:1711.05624. For the tensor-power lift used in Lemma 2.2, any uniform representation with bounded matrix norm already requires $m = \Omega_r(n^{1-1/r})$, even for a hypergraph with one $2r$ -edge. Hence the source choice $m = C_r n^{1-1/r}$ is optimal up to constants. A second theorem gives a polynomial dimension obstruction for the remark's more general nonlinear embedding, but does not settle that broader question.

1 Source and exact question

Jop Briët and Sivakanth Gopi, *Gaussian width bounds with applications to arithmetic progressions in random settings*, arXiv:1711.05624v3 [1], associate to a $2r$ -uniform hypergraph $H = ([n], E)$ the polynomial

$$p_H(x) = \sum_{S \in E} \prod_{i \in S} x_i.$$

Their Lemma 2.2 constructs, for $m = C_r n^{1-1/r}$ and $N = n^m$, a matrix $A \in \mathbb{R}^{N \times N}$ satisfying

$$\|A\| \lesssim_r \Delta(H), \quad p_H(x) = \frac{n}{c_r N} \langle Ax^{\otimes m}, x^{\otimes m} \rangle \quad (x \in \{-1, 1\}^n).$$

Remark 2.3 asks whether the value of m is optimal, then raises a more general question allowing an arbitrary common map into $\mathbb{R}^N$.

The multiplicity of p_H is then exactly the degree $\Delta(H)$. Clearly the coordinate mappings ψ_i of the form featuring in Theorem 1.1 can be written as p_H for some d -hypergraph H of degree at most t . The reduction to the quadratic case is based on the following key lemma, in which for $x \in \mathbb{R}^n$ and $m \in \mathbb{N}$, the m th tensor power x is defined as $x^{\otimes m} = (\prod_{i=1}^m x_{u_i})_{u \in [n]^m}$.

Lemma 2.2 (Matrix lemma). *For every $r \in \mathbb{N}$ there exist a $C_r, c_r \in (0, \infty)$ and $n_0(r) \in \mathbb{N}$ such that the following holds. Let $n \geq n_0(r)$, $m = C_r n^{1-1/r}$ and $N = n^m$. Let $H = ([n], E)$ be a $2r$ -uniform hypergraph and let p_H be the polynomial as in (3). Then, there exists a matrix $A \in \mathbb{R}^{N \times N}$ such that $\|A\| \lesssim_r \Delta(H)$ and for every $x \in \{-1, 1\}^n$,*

$$p_H(x) = \frac{n}{c_r N} \langle Ax^{\otimes m}, x^{\otimes m} \rangle.$$

Moreover, A is the adjacency matrix of a graph (with possible parallel edges).

Remark 2.3. We do not know if the value of m in Lemma 2.2 is optimal. More generally, a better bound in Theorem 1.1 would follow if, for any fixed hypergraphs $H_1, \dots, H_k$ as in the lemma, there exist $N \leq o(n^m)$, matrices $A_1, \dots, A_k \in \mathbb{R}^{N \times N}$ with $\|A_i\| \lesssim_r \Delta(H_i)$ and a map $f : \{-1, 1\}^n \rightarrow \{y \in \mathbb{R}^N : \|y\|_2 \leq \sqrt{N}\}$ such that $p_{H_i}(x) = (n/c_r N) \langle A_i f(x), f(x) \rangle$ for every $i \in [k]$ and $x \in \{-1, 1\}^n$.

Here $x^{\otimes m}$ is indexed by words $u = (u_1, \dots, u_m) \in [n]^m$ and has coordinate $\prod_{j=1}^m x_{u_j}$.

2 Results

Theorem 1 (Optimal tensor-power length). *Fix $r \geq 1$ and positive constants a_r, K_r . Let $S \subset [n]$ have $|S| = 2r$, let $m \geq 1$, and put $N = n^m$. Suppose that a real matrix $A \in \mathbb{R}^{N \times N}$ satisfies $\|A\| \leq K_r$ and*

$$\prod_{i \in S} x_i = \frac{n}{a_r N} \langle Ax^{\otimes m}, x^{\otimes m} \rangle \quad \text{for every } x \in \{-1, 1\}^n. \quad (1)$$

Then

$$m \geq c(r, a_r, K_r) n^{1-1/r}.$$

Corollary 2 (Answer to Remark 2.3, first sentence). *The exponent and asymptotic order of $m = C_r n^{1-1/r}$ in the source's Lemma 2.2 are optimal. This remains true even if the adjacency-matrix requirement is discarded and arbitrary real matrices of norm $O_r(1)$ are allowed.*

The deeper upgrade gives a necessary dimension for the common nonlinear map proposed in the second sentence of the remark.

Theorem 3 (A dimension floor for arbitrary common lifts). *Fix r, a_r, K_r as above. Suppose one map $f : \{-1, 1\}^n \rightarrow \mathbb{R}^N$, with $\|f(x)\|_2 \leq \sqrt{N}$, and matrices A_S with $\|A_S\| \leq K_r$ satisfy*

$$\prod_{i \in S} x_i = \frac{n}{a_r N} \langle A_S f(x), f(x) \rangle$$

for every $x \in \{-1, 1\}^n$ and every $S \in \binom{[n]}{2r}$. Then

$$N \geq c(r, a_r, K_r) n^{2r-2}.$$

3 Proof intuition

Tensor coordinates remember only which symbols occur an odd number of times. Extracting the Fourier coefficient belonging to one edge S therefore produces a matrix that connects two word-parity classes precisely when they differ by S . Its trace norm, normalized by N , is the Hellinger affinity between the parity law of a random word and the law shifted by S .

Hellinger affinity can only increase when all coordinates outside S are forgotten. A word hits S only about $2rm/n$ times. A parity pattern of weight h and its bitwise complement require at least h and $2r - h$ hits, respectively; binomial factorial moments make the resulting affinity $O_r((m/n)^r)$. The source normalization demands a Fourier coefficient of order N/n , forcing $(m/n)^r \gtrsim_r 1/n$.

For a completely arbitrary common lift, the parity description disappears. Instead, Fourier Parseval applied entrywise to $f(x)f(x)^\top$ gives a weaker but still nontrivial polynomial dimension lower bound.

4 Parity-affinity bound

For a word $u \in [n]^m$, let $\pi(u) \subset [n]$ be the set of symbols occurring an odd number of times. If U is uniform on $[n]^m$, write

$$\mu(\alpha) = \Pr[\pi(U) = \alpha].$$

For $S \subset [n]$, define the $N \times N$ zero-one matrix

$$P_S(u, v) = \mathbf{1}_{\{\pi(u) \Delta \pi(v) = S\}}.$$

Lemma 4 (Trace norm as parity affinity). *For every S ,*

$$\frac{\|P_S\|_*}{N} = \sum_{\alpha \subset [n]} \sqrt{\mu(\alpha)\mu(\alpha \Delta S)}.$$

If $|S| = 2r$ and $m \leq n/(4r)$, the right-hand side is at most $C_r(m/n)^r$.

Proof. Let $c_\alpha = N\mu(\alpha)$ be the number of words with parity α . After rows and columns are grouped by parity, P_S has, for each α , one all-ones block of size $c_\alpha \times c_{\alpha \Delta S}$, and these blocks have disjoint row and column sets. Each block has its unique nonzero singular value $\sqrt{c_\alpha c_{\alpha \Delta S}}$. Summing proves the identity.

The displayed sum is the Hellinger affinity between μ and its translate by S . Group its terms according to the restriction $z = \alpha \cap S$. Cauchy–Schwarz within every group shows that forgetting coordinates outside S can only increase affinity. Thus it is at most

$$\sum_{z \subset S} \sqrt{\Pr[Z = z] \Pr[Z = S \setminus z]}, \quad (2)$$

where $Z = \pi(U) \cap S$.

Let L count positions of U whose symbol lies in S . Then $L \sim \text{Bin}(m, 2r/n)$. If $|z| = h$, the event $Z = z$ requires $L \geq h$, while $Z = S \setminus z$ requires $L \geq 2r - h$. For every integer $j \geq 1$, the factorial-moment bound gives

$$\Pr[L \geq j] \leq \mathbb{E} \binom{L}{j} = \binom{m}{j} (2r/n)^j \leq \frac{(2rm/n)^j}{j!};$$

the corresponding bound for $j = 0$ is 1. Hence every summand in (2) is at most $C_r(m/n)^r$. There are only 2^{2r} summands, completing the proof. $\square$

5 Proof of tensor optimality

Proof of Theorem 1. Replacing A by its symmetric part changes neither the quadratic form nor the norm bound. Put $q_A(x) = \langle Ax^{\otimes m}, x^{\otimes m} \rangle$. Fourier orthogonality on $\{-1, 1\}^n$ gives

$$\begin{aligned} \widehat{q}_A(S) &= 2^{-n} \sum_{x \in \{-1, 1\}^n} q_A(x) \prod_{i \in S} x_i \\ &= \sum_{u, v \in [n]^m} A_{uv} \mathbf{1}_{\{\pi(u) \Delta \pi(v) = S\}} = \text{tr}(A^\top P_S). \end{aligned}$$

By operator/trace-norm duality and Lemma 4,

$$|\widehat{q}_A(S)| \leq \|A\| \|P_S\|_* \leq K_r C_r N (m/n)^r \quad (3)$$

whenever $m \leq n/(4r)$.

On the other hand, (1) says $q_A = (a_r N/n) \prod_{i \in S} x_i$, so $|\widehat{q}_A(S)| = a_r N/n$. Comparing with (3) yields

$$m^r \geq \frac{a_r}{K_r C_r} n^{r-1}.$$

If $m > n/(4r)$, the desired lower bound is immediate after decreasing the constant. This proves the theorem in all cases. $\square$

6 The arbitrary-lift dimension bound

Proof of Theorem 3. Set $F(x) = f(x)f(x)^\top$ and define its matrix-valued Fourier coefficients by

$$\widehat{F}(T) = 2^{-n} \sum_{x \in \{-1,1\}^n} \left(\prod_{i \in T} x_i \right) F(x).$$

The assumed representation gives

$$\text{tr}(A_S^\top \widehat{F}(S)) = a_r N/n.$$

Schatten duality followed by $\|M\|_* \leq \sqrt{N}\|M\|_F$ therefore implies

$$\|\widehat{F}(S)\|_F^2 \geq \frac{a_r^2 N}{K_r^2 n^2}.$$

Entrywise Fourier Parseval and $\|F(x)\|_F = \|f(x)\|_2^2$ give

$$\sum_{T \subset [n]} \|\widehat{F}(T)\|_F^2 = \mathbb{E} \|F(x)\|_F^2 = \mathbb{E} \|f(x)\|_2^4 \leq N^2.$$

Summing the preceding lower bound over all $\binom{n}{2r}$ choices of S yields

$$N \geq \frac{a_r^2}{K_r^2 n^2} \binom{n}{2r} \geq c(r, a_r, K_r) n^{2r-2}$$

for all sufficiently large n ; finitely many smaller n are absorbed into the constant. $\square$

7 Scope, verification, and novelty

The first theorem answers the tensor-power optimality question at the natural asymptotic-order level. It does not optimize the constants C_r, c_r . The second theorem does not rule out the $N = o(n^m)$ lift contemplated in the more general sentence of Remark 2.3, since its lower bound is only polynomial. Consequently, the improved Gaussian-width exponent and the linked coding and arithmetic-progression conjectures remain open.

The proof was checked independently through its Fourier normalization, block singular values, Hellinger marginalization direction, factorial-moment tails, and Parseval constants. The supplied script performs small exact parity-count and numerical trace-norm spot checks.

Bounded searches used the source arXiv id and title, the exact Remark 2.3 wording, “matrix lemma”, tensor-power/parity/Hellinger terms, Gaussian-width improvements, and later work on three-query locally decodable codes. They found the source and continuing broader LDC work, but no claimed answer to the isolated m -optimality question. Novelty confidence is moderate; independent expert literature review and proof verification are required.

References

- [1] J. Briët and S. Gopi, *Gaussian width bounds with applications to arithmetic progressions in random settings*, International Mathematics Research Notices 2020, no. 22, 8673–8698; arXiv:1711.05624v3.